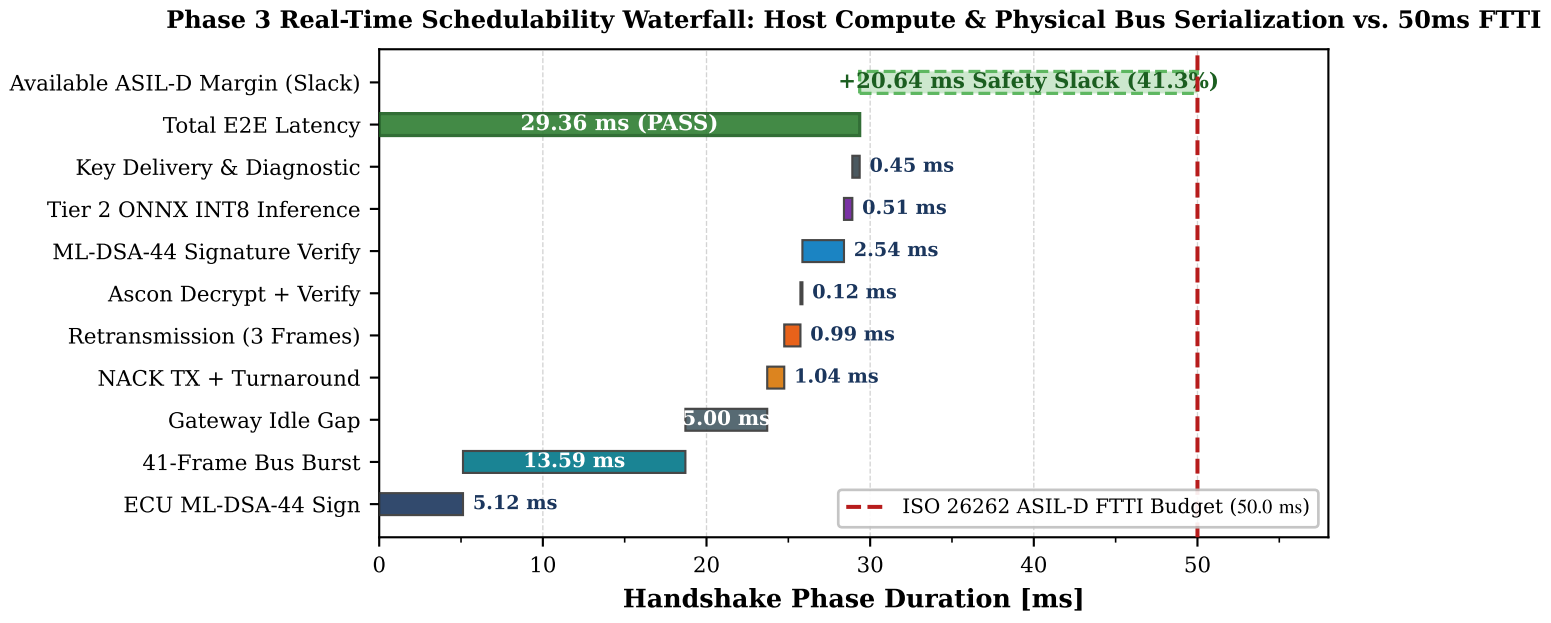
Phase 3 Real-Time Schedulability Waterfall: Host Compute & Physical Bus Serialization vs. 50ms FTTI
Available ASIL-D Margin (Slack)
Total E2E Latency
Key Delivery & Diagnostic
Tier 2 ONNX INT8 Inference
ML-DSA-44 Signature Verify
Ascon Decrypt + Verify
Retransmission (3 Frames)
NACK TX + Turnaround
Gateway Idle Gap
41-Frame Bus Burst
ECU ML-DSA-44 Sign
+20.64 ms Safety Slack (41.3%)
29.36 ms (PASS)
0.45 ms
0.51 ms
2.54 ms
0.12 ms
0.99 ms
1.04 ms
5.00 ms
13.59 ms
5.12 ms
ISO 26262 ASIL-D FTTI Budget (50.0 ms)
0
10
20
30
40
50
Handshake Phase Duration [ms]